

Stacks

(Assignment Questions)

Question 1 : Given the head of a singly linked list, return true if it is a Palindrome or false otherwise. [\[Go to Qs\]](#)

Examples :

Input: head = [1,2,2,1]

Output: true

Input: head = [1,2,3]

Output: false

Note - Use a stack to solve this problem.

Question 2 : Given an encoded string, return its decoded string.

The encoding rule is: $k[\text{encoded_string}]$, where the `encoded_string` inside the square brackets is being repeated exactly k times. Note that k is guaranteed to be a positive integer.

You may assume that the input string is always valid; there are no extra white spaces, square brackets are well-formed, etc. Furthermore, you may assume that the original data does not contain any digits and that digits are only for those repeat numbers, k . For example, there will not be input like `3a` or `2[4]`. [[Go to Qs](#)]

Examples :

Input: s = "3[a]2[bc]"

Output: "aaabcbc"

Input: s = "2[abc]3[cd]ef"

Output: "abccabccdcddcdef"

Question 3 : We have an absolute path for a file(Unix-style), simplify it. Note that absolute path always begins with '/'(rootdirectory), a dot in path represents current directory and double dot represents parent directory.[[Go to Qs](#)]

Examples :

Input: path = "/home//foo/"

Output: "/home/foo"

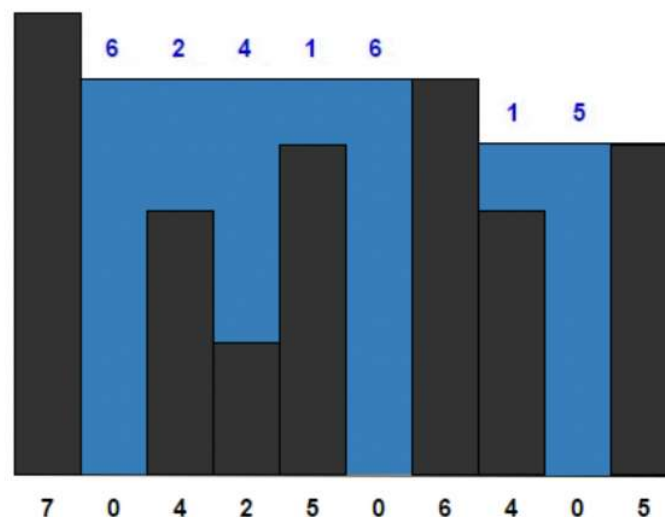
Explanation: Multiple consecutive slashes are replaced by a single one.

Input: path = "/home/user/Documents/../Pictures"

Output: "/home/user/Pictures"

Explanation: A double period ".." refers to the directory up a level.

Question 4 : Given n non-negative integers representing an elevation map where the width of each bar is 1, compute how much water it can trap after raining.[[Go to Qs](#)]



Examples :

Input: height = [7, 0, 4, 2, 5, 0, 6, 4, 0, 5]

Output: 25

Explanation: The above elevation map (black section) is represented by the array [7, 0, 4, 2, 5, 0, 6, 4, 0, 5]. In this case, 25 units of rain water (blue section) are being trapped.